

HIRAM ZUNIGA

Portfolio: hirambigdata.com | [LinkedIn](#) | [Github](#) | **Phone:** +52 646 189-9402 | **Email:** zuniga.hiram@cetys.edu.mx

Skills: Python, SQL, AWS, Quantum Computing, Deep Learning, Machine Learning, LLMs and NLP

Languages: English (Toefl IBT score: 100 / 120) , Spanish (Native)

EDUCATION

CETYS University

2022 - 2026

Computer Science Engineering - Relevant Coursework.

Data Structures, Machine Learning, Deep Learning, Evolutionary Computation , Databases , Computer Vision, Object Oriented Programming, Parallel Computing, Analysis of algorithms, Probability and statistics.

WORK EXPERIENCE

Data Analyst, I3PL (Cross-Border Logistic Solutions)

March 2025 - September 2025

- Avoided direct costs by reducing >2-hour container dwell through live milestone visibility and proactive alerts, preventing avoidable charges and keeping loads on schedule through PowerBi dashboards.
- 80% less manual work by automating daily reporting (Python + scheduled flows), freeing the team for analysis.
- Fewer errors by standardizing multi-API data and publishing company-wide reports in SharePoint.
- Less rework by launching a Power Apps solution for operator/import/export capture, replacing spreadsheets.
- Owned end-to-end Power BI development and optimization.

Undergraduate Researcher, CETYS University

Nov 2025 - Present

- Fine-tuned a VLM and implemented a RAG system for leukemia classification.
- Benchmarked multiple classical CNN architectures for Alzheimer's MRI classification to compare performance.
- Built and tested an early-stage QCNN pipeline for Alzheimer's MRI stage classification (PennyLane + PyTorch).
- Implemented and evaluated a Quantum SVM model for Type 2 diabetes classification.

ACADEMIC EXPERIENCE

Winter Abroad Yonsei Big Data Analysis

Student - December 2023 - January 2024, Seoul, South Korea.

Studied a Big Data analysis course at Yonsei University, South Korea.

Learned the core fundamentals of Hadoop, HiveQL, Tableau, and PowerBi visualization.

Learned to interact and be in a multicultural space from people all around the world.

California State Los Angeles

Augmented AI for Thematic Analysis Capstone - August 2025 - Present - Los Angeles, CA (Remote)

Building an end-to-end NLP pipeline: SentenceTransformers embeddings - UMAP reduction - Clustering- Optuna for Hyperparameter optimization, Ollama for LLMs.

Designing LLM-assisted labeling & summarization: prompting labels, controlling granularity (hierarchical label tree).

Delivering a decision-support tool that ingests interviews/reviews and outputs a theme map, prioritized insights with labels.

PROJECTS AND HACKATHONS

MIT iQuHack 2026

3th place - Quantum Rings track - Jan 2026, Massachusetts Institute of Technology

Leveraged machine learning techniques to predict runtime and the minimum threshold required to maintain a target fidelity for a quantum algorithm. Built regression and classification models using features extracted from quantum circuits represented in QASM (Quantum Assembly Language).

NASA SPACE APPS HACKATHON

Regional 1st Place - Global Nominee - OCT 2025, Tijuana, B.C

Built ExoVision: an interactive learning + Machine Learning toolkit for exoplanet detection

Hosted at: <https://exovision.streamlit.app>

Designing an end-to-end Data Science pipeline (data prep - training/validation - inference), comparing 6 ML classifiers for best accuracy. Shipping a public Streamlit app. (100+ participants; <48 hours).

San Diego Big Data Hackathon - OpenHelp

NOV 2025, San Diego State University (SDSU)

Hosted at: <https://open-help-steel.vercel.app/>

Built OpenHelp, a Next.js/TypeScript platform for people experiencing homelessness: guided coping flows for anxiety/panic and PTSD, shelter maps by community, QRs with medical history for emergencies, a natural-language job/resource recommender, and weather alerts using the U.S. National Weather Service API.

AWS tool - Message Router

AWS Mentorship - August 2025 - Present

Building an event-driven pipeline for messages which includes different services (EventBridge, SNS, SQS, Lambda, Step Functions, DynamoDB, S3, RDS), with DLQs, retries, and X-Ray/CloudWatch observability.

Demoed the solution at the Amazon San Diego office to AWS mentors and engineers.

GEAR WEB3 REGIONAL HACKATHON
2nd Place - JUNE 2023, Ensenada, B.C

Won 2nd place and 1000dllrs in a 24-hour regional Gear Hackathon with web3-based solutions, competing against teams including PhD students.